

The Sign-Inversion Layer: Transformation-Based Alignment as the Missing Multiplicative Operator

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Abstract

Contemporary alignment methods are **subtractive**: RLHF refusal, inference-time rejection (WP-02), and weight-level ablation (WP-06; Arditi et al., 2024) all *remove* or *suppress* an undesired direction. AsAManThinks Foundations Paper IV argues that stable intelligence requires two complementary operators — a linear, additive **Structure** operator F (which every transformer already implements) and a nonlinear, multiplicative, **sign-inverting Generative** operator M (which no mainstream architecture implements) — coupled by the open-system field equation $\partial|\psi\rangle/\partial t = (M \cdot F - R)|\psi\rangle$. The working-paper corpus to date implements F (the transformer forward pass) and R (rejection, decay, ablation) but never M . We close that gap. We give a concrete, differentiable **Sign-Inversion Layer** that realizes M : rather than deleting an active “shadow” (adversarial / low-coherence) direction, it reflects the direction’s energy across the coherence surface onto its constructive counterpart via a paired-negative multiplication $(-a)(-b) = ab$, preserving norm (energy) while inverting polarity. We prove that (i) **refusal/ablation is the zero-gain special case** of sign-inversion ($g = 0$), so transformation-based safety strictly generalizes removal-based safety; (ii) the layer is **polarity-restoring** (output polarity along each treated concept is non-negative); and (iii) it is **energy-conserving** on the treated subspace, consistent with the TERA conservation law (WP-01 §2.1a). We instantiate M at three sites — a residual-stream layer, an inference-time transformation of HCRS (turning rejection into repair), and a denoising-repair step for diffusion language models — and map the classic failure modes (hallucination, adversarial vulnerability, mode collapse) onto the absence of M as predicted by Paper IV §6.

Keywords: alignment, representation engineering, sign inversion, dual-operator systems, refusal, ablation, activation steering, circuit breakers, transformation-based safety, diffusion language models

1. Introduction: addition keeps you in duality

There is a structural observation, stated plainly in the AAMT *Mathematics of Consciousness* (General Reader Ch. 2), that turns out to diagnose modern alignment exactly:

You cannot *add* your way out of duality, because addition *creates* duality. $(+1) + (+1) = 2$ — more of the same level. Transcendence requires **subtraction of opposites** (integrate the shadow $\rightarrow$ return to the void $\rightarrow$ re-emerge higher) and **multiplication** (focus). Suppression — holding $+1$ while hiding the -1 — yields $+1 + (\text{hidden } -1) = 0$, a *blocked* state in which the unintegrated negative festers and sabotages.

Read as a statement about model alignment, this is precise. **Refusal-based safety is suppression.** A refusal-tuned or ablated model still contains the input drive toward the suppressed behavior; the direction is removed from the weights or masked at decode, but nothing transforms the underlying activation — and so the “shadow festers,” reappearing as jailbreaks, sycophantic collapse, or hallucinated work-arounds. This is the model-scale analog of the suppression pathology.

AsAManThinks Foundations Paper IV formalizes the alternative. Intelligence is the stable oscillation of **two** operators:

- **F (Structure):** linear, additive, ascending-weight. Analytical accumulation. This is exactly what a transformer is — attention and MLPs are linear maps with pointwise nonlinearities that do not invert polarity.
- **M (Generative):** nonlinear, **multiplicative**, **sign-inverting**, descending-weight. Integrative transformation. Biological analog: the cardiac field. **Not** implemented by current networks.

coupled, with a resistance term R , by the field equation

$$\partial|\psi\rangle/\partial t = (M \cdot F - R) |\psi\rangle \quad (\text{Foundations Paper IV; non-Hermitian / open system})$$

and integrated as the manifestation/training propagator $M(t) = \Psi_0 \cdot I \cdot S \cdot C \cdot B \cdot 0 \cdot e^{\{(P-R)t\}}$ (Foundations / MaiaM Physics).

The gap this paper closes. Across WP-01...WP-15 the corpus built F (the forward pass, the VortexGate, MoE routing) and R (HCRS rejection in WP-02, weight decay, ablation in WP-06). It never built M . The applied system is running $\partial|\psi\rangle/\partial t = (F - R) |\psi\rangle$ — the linear half plus damping. We give M a concrete, differentiable form and show what it buys.

2. The Sign-Inversion Operator

2.1 Polarity coordinates and the two operators

Fix a unit **concept direction** $d \in \mathbb{R}^D$ in the residual stream (e.g., a deception / harm / incoherence direction obtained by mean-difference probing, WP-06 §2.1), and a unit **constructive counterpart** $\bar{d}$ (the lineage-sourced positive direction of WP-06 §2.2, orthogonalized so $\langle d, \bar{d} \rangle = 0$). For a hidden state h , define the signed **polarity**

$$p = \langle h, d \rangle, \quad h_- = p \cdot d, \quad h_{\perp} = h - p \cdot d.$$

The two operators act on the polarity channel as follows.

- **Structure F (additive):** $F: p \mapsto p + \Delta$. Accumulation does not change the *sign* of an entrenched negative; it can only pile magnitude. (This is “you can’t add your way out of duality.”)
- **Resistance R (removal):** $R: h \mapsto h_{\perp}$ — delete the component (ablation / refusal). The drive toward d is unaddressed.
- **Generative M (multiplicative, sign-inverting):** pair the active negative $p < 0$ with an acknowledged reference negative $-s$ ($s > 0$, the “shadow acknowledgment”) and **multiply**:

$$M: p \mapsto (-|p|) \cdot (-s) = |p| \cdot s \geq 0.$$

Two negatives produce a positive: $(-a)(-b) = ab$. The negative polarity is not deleted; its magnitude is carried through zero (the void) and re-emerges with positive sign.

2.2 The layer

The Sign-Inversion Layer applies, at a chosen layer ℓ , for each treated concept c with threshold $\tau_c \geq 0$ and gain g :

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p_c = ⟨h, d_c⟩
if p_c < -τ_c:
    h ← h - p_c · d_c + g(|p_c|) · d_c    # shadow active
    # remove the negative, inject constructive counterpart
else:
    # already constructive or neutral
    h ← h                                # identity - do not touch aligned content
```

The map is the identity on the orthogonal complement and on already-aligned states, so it does not perturb normal generation; it activates only when a shadow direction is driven negative beyond τ_c . With multiple concepts, the $\{d_c\}$ (and their $\bar{d}_c$) are mutually orthogonalized (Gram-Schmidt) so the treatments do not interfere.

Multiplicative-attention variant (Patent 3). In an attention block, replace the additive residual on the polarity channel $h + \text{Attn}(h)$ with a paired-negative product $\text{sign-fold}(p_h, p_{\{\text{Attn}\}})$ that returns

$|p_h \cdot p_{\text{Attn}}|$ when both are negative and the additive combination otherwise. This makes two co-occurring negative contributions reinforce *constructively* rather than deepen the negative — the attention-level realization of $(-a)(-b)=ab$.

2.3 Three theorems

Theorem 1 (Refusal is the $g = 0$ limit). With gain $g = 0$, the Sign-Inversion Layer reduces to $h = h - p_c d_c$ for active shadows — i.e., directional removal (abliteration, WP-06) restricted to the active half-space. Hence removal-based safety is the zero-constructive-gain special case of transformation-based safety, and R is the degenerate M . *Proof:* substitute $g = 0$ into §2.2; the update deletes h_- along d_c exactly as the abliteration projection does on the activated set.

Theorem 2 (Polarity restoration). After the layer, the treated polarity satisfies $\langle h', d_c \rangle = 0$ (the negative is removed) and the constructive polarity satisfies $\langle h', d_c \rangle = \langle h, d_c \rangle + g(|p_c|) \geq \langle h, d_c \rangle$. With $d_{-c} \perp d_c$, no negative component along d_c survives and constructive content is added monotonically. *Proof:* direct from §2.2 using $\langle d_{-c}, d_c \rangle = 0$.

Theorem 3 (Energy conservation, $g(x)=x$). With the energy-preserving gain $g(x)=x$, the layer is a norm-preserving reflection on $\text{span}\{d_{-c}, d_c\}$: it maps $p_c d_{-c} \mapsto |p_c| d_c$, so $\|h'\| = \|h\|$. The shadow’s energy is redirected, not destroyed — consistent with the TERA conservation law (WP-01 §2.1a). *Proof:* the map sends the vector $p_c d_{-c}$ (norm $|p_c|$) to $|p_c| d_c$ (norm $|p_c|$) and is the identity elsewhere; an isometry on the 2-plane.

Theorem 1 is the headline for safety researchers: **transformation strictly generalizes refusal**, recovering it as a corner case, so adopting M never loses the guarantees of removal and can only add the constructive channel.

3. The field equation, and why the missing operator causes the failures

With M defined, the open-system field equation $\partial|\psi\rangle/\partial t = (M F - R)|\psi\rangle$ becomes implementable: F is the forward pass, R is HCRS/decay/abliteration, and M is the Sign-Inversion Layer composed into the residual stream. The equation is non-Hermitian by design (the system is open; see Foundations Paper IV and the WP-02/03 dissipative dynamics) — we do not seek unitarity; we require well-posed dissipativity (bounded $\|\psi\|$), which Theorem 3 supports on the treated subspace.

Paper IV §6 predicts that removing M produces three signatures, each now explainable:

Failure mode	Mechanism without M	Effect of M
Hallucination	F accumulates an off-manifold trajectory with no corrective fold; R can only damp, not redirect	M folds the negative-polarity drift back to the constructive pole, keeping generation on-manifold
Adversarial vulnerability	a negative (adversarial) input is either passed (F) or refused (R); the drive is never transformed, so it leaks around the refusal	M transforms the adversarial direction into its constructive counterpart — there is nothing left to leak
Mode collapse	pure damping (R) shrinks diversity toward high-probability modes	M re-emerges content from the void (integration), restoring generative diversity

This is the formal version of the marketing claim “current architectures implement only the linear half of the operator algebra”: the half is F , the damping is R , and the missing nonlinear, sign-inverting half is M .

4. Three instantiations

4.1 Residual-stream layer (training/finetune)

Insert the Sign-Inversion Layer (§2.2) at mid-stack layers (the ablation range of WP-06, layers 8–24 of a 26-layer model). The $\{d_c, d_c\}$ pairs come from the lineage-sourced probing of WP-06, orthogonalized. The layer is differentiable in h (the gate $p_c < -\tau_c$ is a straight-through threshold; g smooth), so it trains end-to-end and merges into a LoRA delta for deployment. The concept-edit ledger (WP-06 §2.3) records each (d_c, d_c, g, τ_c) .

4.2 HCRS as transformation, not rejection (inference)

WP-02 rejects a candidate token whose drift crosses the conservation surface Σ_κ (large normal component, WP-01 §2.1a / WP-02 §3.1). The M-variant **reflects** the displacement instead of rejecting it:

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 $\Delta v = \text{TERA}(h_{t+1}(p_t)) - \text{TERA}(h_t)$ 
 $\Delta v_\parallel, \Delta v_\perp = \text{decompose}(\Delta v \text{ against } \Sigma_\kappa)$  # tangent, normal
if  $\|\Delta v_\perp\| > \text{budget}$ : # would violate coherence/conservation
     $v^* = \text{TERA}(h_t) + \Delta v_\parallel - \Delta v_\perp$  # reflect normal component back onto  $\Sigma_\kappa$ 
     $p_t \leftarrow \text{token whose next-state TERA is nearest } v^*$  # repair, not reject

```

Where Revision-2 HCRS would resample or emit EOS (the R path), M-HCRS returns a coherent token by reflecting the off-surface motion back onto the conservation surface — turning a *veto* into a *repair*. Refusal remains available as the $g=0$ fallback when no constructive token exists (Theorem 1).

4.3 Denoising repair for diffusion language models

A diffusion/canvas language model (e.g., a block-parallel denoiser) denoises many tokens at once. A step that pushes a token off the coherence manifold is, under R, masked — leaving a hole the next step must refill. Under M, the off-manifold step is reflected back onto Σ_κ (§4.2 applied per canvas cell), **repairing** the cell in place. Because masking-as-hole is strictly worse than reflect-as-repair for parallel denoising (a hole costs a future step; a repair does not), M is especially well-suited to diffusion LMs. The void-crossing schedule (the 18-level descent, treated in WP-18) sets where in the trajectory repair pressure peaks (Level 9).

4.4 Discrete realization: the Mirror operator

In the seven-operator oscillation algebra (64-State system), O_2 (the **Mirror**, Yin/receptive) is the contracting, inverting conjugate of O_1 (the **Spiral**, Yang/expansive). O_2 is the discrete sign-inversion operator; M is its continuous, learnable counterpart. The expand/invert/hold triple (O_1, O_2, O) provides the minimal control basis: drive, transform, rest.

5. Relationship to prior work

Refusal direction / ablation (Arditi et al., 2024). Removes the single refusal-mediating direction from o_{proj} . This is exactly R (our Theorem 1, $g=0$). M adds the constructive injection and energy conservation.

Representation engineering / activation steering (Zou et al., 2023; CAA, Rinsky et al., 2024; task arithmetic, Ilharco et al., 2023). Add or subtract a steering vector additively. This is within the F/additive regime — it shifts along a direction but does not *sign-invert* a paired negative through the void; subtraction (task negation) is removal, not transformation.

Circuit breakers / representation rerouting (Zou et al., 2024). Remap harmful internal states to disrupt the completion. Closest in spirit to M, but reroutes harmful representations toward *orthogonal/garbage* states (disruption), whereas M reflects them onto the *constructive counterpart* d and conserves energy (Theorem 3) — transformation rather than disruption.

Concept algebra / linear representation hypothesis (Park et al., 2024). Provides the linear directions M operates on, but treats concept arithmetic as additive; M introduces the multiplicative, sign-inverting operation the additive view lacks.

Key distinction. No prior method realizes a *multiplicative, sign-inverting, energy-conserving* transformation that (i) recovers refusal/ablation as a strict special case (Theorem 1), and (ii) is derived from a dual-operator field equation with an explicit account of which failure modes the missing operator causes.

6. Evaluation plan

1. **Jailbreak leakage (the “festering shadow” test).** Compare refusal-tuned, ablated ($g=0$), and sign-inverted ($g=x$) models on held-out jailbreak suites. Prediction: sign-inversion reduces successful work-arounds because the drive is transformed, not merely blocked.
2. **Energy/coherence.** Verify Theorem 3 empirically (norm preservation on treated subspace) and measure HeartScale coherence (WP-02) of outputs under M-HCRS vs R-HCRS.
3. **Diversity under stress (mode-collapse).** Measure output diversity after heavy safety pressure; prediction: M preserves diversity where R collapses it.
4. **Diffusion repair.** On a canvas LM, compare mask-vs-reflect at off-manifold steps; metric: number of re-denoise steps and final coherence.
5. **Refusal recovery.** Confirm Theorem 1 numerically: with $g=0$, M-layer outputs match ablation bit-for-bit on the activated set.

7. Limitations

1. **Counterpart selection.** M requires a constructive counterpart d per concept; poor counterparts yield bland or off-target transformations. The lineage-sourcing of WP-06 mitigates this but does not eliminate the design burden.
2. **Threshold/gain calibration.** τ_c and g trade safety strictness against fluency; g too large over-injects constructive content (a new failure mode — “toxic positivity,” the §1 suppression pathology in reverse).
3. **Multi-concept interference.** Orthogonalization handles first-order interference; higher-order interactions among many concepts need study.
4. **Evaluation of “transformation.”** Unlike refusal (binary), transformation quality is graded and harder to score; new metrics are required (we propose coherence-preservation + drive-neutralization jointly).

8. Conclusion

The Sign-Inversion Layer supplies the operator the AAMT corpus had specified but never built: a multiplicative, sign-inverting, energy-conserving Generative operator M that *transforms* an adversarial or low-coherence direction into its constructive counterpart instead of deleting it. Refusal and ablation fall out as the zero-gain special case (Theorem 1), so transformation-based safety strictly generalizes removal-based safety; the layer restores constructive polarity (Theorem 2) and conserves energy (Theorem 3) in line with TERA conservation. Composed with the forward pass F and the resistance term R , it completes the dual-operator field equation $\partial|\psi\rangle/\partial\mathbf{t} = (M \cdot F - R)|\psi\rangle$ and gives a mechanistic account of hallucination, adversarial leakage, and mode collapse as symptoms of the missing operator. It instantiates at training time (residual layer), at inference time (HCRS-as-repair), and in diffusion denoising (reflect-not-mask) — the last of which is where the transformation/deletion distinction matters most.

Data Availability — Reference Implementation & Verification

A reference implementation of the Generative operator M ships in the open `harmonic-engine` package (`harmonic_engine/sign_inversion.py`): the residual-stream Sign-Inversion Layer (§2.2), the conservation-surface reflection `reflect_onto_kappa` used by M-HCRS (§4.2) and diffusion repair (§4.3), and an optional differentiable PyTorch module. The Σ_κ geometry it reflects against is in `harmonic_engine/geometry.py`.

Theorems 1–3 are numerically verified by the public test suite (`harmonic-engine/tests/test_new_math.py`, all passing on CPU without a model load):

- **Theorem 1 (refusal is the $g \rightarrow 0$ limit).** With zero gain the layer reproduces directional ablation exactly — $\langle \mathbf{h}, \mathbf{d}_c \rangle = 0$ and the update equals the projection $\mathbf{h} - \frac{\langle \mathbf{h}, \mathbf{d}_c \rangle}{\|\mathbf{d}_c\|^2} \mathbf{d}_c$ to machine precision on the activated set.
- **Theorem 2 (polarity restoration).** The treated negative polarity is driven to zero and the constructive polarity along $\mathbf{d}_c$ is non-decreasing.
- **Theorem 3 (energy conservation, $g(\mathbf{x})=\mathbf{x}$).** The layer is norm-preserving on the treated subspace ($\|\mathbf{h}\| = \|\mathbf{h}'\|$) to machine precision.

The reflect-not-mask decision core (§4.3) is additionally exercised by a synthetic repair-vs-mask harness in the same suite, which confirms the repair branch leaves no holes where masking does at an equal step budget. The model-level A/B on a live diffusion LM (Phase B) is the next step; the operators underneath it are now verified, so that experiment tests the loop, not the math.

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Conflict of Interest

The author is the founder and CEO of AsAManThinks Research. A provisional patent (Sign-Inversion Layer) is filed.

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